



Examination	: B.Tech Semester v	Seat No.	:
Date	: 1-8-2022	Day	: Monday
Time	: 12:30 to 2:45 PM	Max. Marks	: 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed.

- CO2 (a) If [AX]=F000H, after the execution of the instruction CBW, the contents of AX would be ____ and if AX=0F85H then after execution of the instruction CWD, the content of DX is _____. [12]
- CO1 (b) State whether the given instructions are valid if not then correct them. [2]
- i. MUL 4
 - ii. DEC [SI]
- CO1 (c) What are the differences between MOV SI, offset Array and LEA SI, Array instructions? [2]
- CO1 (d) What do you mean by segment override prefix? Name the control flags and their working. [2]
- CO2 (e) If [BX]=0050H, [DS]=2000H, [00500H]=80H, [20050H]=08H, [CX]=5000H, then after MOV CL,[BX] instruction is executed the contents of CX will be _____. [2]
- CO3 (f) What happens when a high is applied to RESET pin? [1]
- CO2 (g) What are the function of ALE, DEN' pins of 8086 [1]

Q.2 Attempt Any TWO from the following questions.

- CO2 (a) Design 8086's memory system consisting of 512K bytes of RAM memory and 128K bytes of ROM using 128K bytes of RAM and 64K bytes of ROM Chip. RAM memory is to reside from the address 00000H and the address of the ROM is to be from A0000H. Use Full decoding. [6]
- CO2 (b) Using functions of INT 21h read a character through keyboard. If the character 'Y' is pressed then it should display the message "Yes" or else for any other key it has to display "No". [6]
- Function no. to read a character is 01 and to display string is 09
- CO2 (c) Explain and Draw the interface required for demultiplexing address/data pins of 8086 processor. [6]

Q.3 Attempt Any ONE from the following questions.

- CO2 (a) Write an assembly language program to find the Maximum of an array of words. Maximum number should be stored at offset address 2000H. [6]
- CO1 (b) Draw and Explain the timing diagram of Memory READ cycle. [6]
- OR
- CO2 (a) Write an Assembly Language program to find whether a given number is prime. If it is a prime number store 01 or else store FFH at offset 2000H of data segment. [6]
- CO1 (b) i. Explain various addressing modes supported by 8086 processor with examples. [4]
- ii. Compare NEAR and SHORT jump of 8086 processor. [2]

CO1: To introduce students with the architecture and operation of typical microprocessors and microcontroller.
CO2: To familiarize the students with the programming and interfacing of microprocessors.
CO3: To provide foundation for designing real world applications using microprocessors and microcontroller.